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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours)

REGULAR EXAMINATION OCTOBER 2026

FIFTH SEMESTER

Discipline Specific Core Course (DSC) – MG5DSCJMC301 – DATA JOURNALISM

(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 70

PART A

Multiple Choice Questions

Answer all the questions

Each question carries **1** mark

1. Option D
2. Option A
3. Option D
4. Option D
5. Option C
6. Option C
7. Option C
8. Option D
9. Option A
10. Option B

[1x10=10]

PART B

Short Answer Type Questions

Answer any **10** questions

Each question carries **3** marks

11. Importance of data – data analysis – data cleaning – data can give journalists leads on the topic – analysis of data across different time periods – data journalism in investigative reporting – establishing connections from collected data – common links between datasets

12.

Descriptive Statistics: Descriptive statistics summarise and describe the main characteristics of a dataset. They help journalists present large amounts of information in a simple and understandable form.

Percentages and Proportions: Percentages express a value as a part of 100. They allow journalists to explain the relative size of a group or the share of a particular category within a total.

Ratios and Rates: Ratios compare two quantities, while rates measure how frequently something occurs in relation to a population, period, or other specified unit.

Frequency Distribution: Frequency distribution shows how often particular values or categories appear in a dataset. It organises information into categories, making patterns easier to identify.

Data comparison – Trend Analysis – Correlation – Sampling and Survey Analysis – Percentage Change – Cross Tabulation – Data Visualization

13. Quantitative Data – Qualitative Data – Primary Data -Secondary Data – Structured Data – Unstructured Data – Timeseries Data – Geospatial Data – Demographic Data – Financial and Economic Data
14. Define of RTI – Provisions in RTI – RTI as an information fetching tool – RTI as a tool to check accountability – Requesting large datasets via government portals, data in the public domain
15. Volume – Quantity of data fetched or given – Measuring units of data
16. Data organization is the process of structuring, ordering and arranging the data in desired format as per requirements. For different kinds of reports, different forms of data organization will be required. Sometimes we need tabular data, sometimes ascending/descending order etc.
17. Clarity, accuracy, simplicity, and consistency.
18. Selection can be based on the following criteria – Type of data, purpose of the story, target audience, ease of use, interactivity, and cost and compatibility.
19. Useful information can be extracted from a PDF report through PDF scraping. **PDF scraping** is the process of extracting text, tables, images, or other useful information from a PDF (Portable Document Format) file for analysis or further use. This process may include the following steps: Identify the PDf, extract the data, clean the data, organize the data and analyze the data.
20. Web scraping is the process of automatically collecting information from multiple web pages using software or scripts. To handle multiple web pages in this process, one should follow the following steps: Identify the websites, inspect the web pages, choose a scraping tool, extract data, handle multiple web pages (use loops, pagination or links to navigate through multiple pages), clean and organize the data, and verify the data.

21. A journalist can identify misleading patterns in a dataset by carefully examining the data, checking its accuracy, and looking for distortions or inconsistencies. One can follow the following steps to ensure the same: check the data source, look for missing data, identify outliers, check for bias, verify statistical claims, compare with other sources, examine visualization techniques, and distinguish correlation from causation.
22. May be written as per the skill of the candidate
    Include the following details:
    Genre of the story – Subject of the story – Basic details of the story including statistics if any – Types of data that would come in the story – Types of data sources you would select (primary/secondary/digital sources and so on) and state the reasons for the selection.

[3x10=30]

PART C

Essay Type Questions

Answer any **2** questions

Each question carries **15** marks

23. Data journalism is the practice of collecting, analyzing, interpreting, and presenting data to discover and communicate news stories.
    Major concepts:
    Data Collection: Gathering information from sources such as government reports, surveys, websites, public records, and databases.

    Data Cleaning: Removing errors, duplicates, and inconsistencies to improve the quality of data.

    Data Analysis: Examining data to identify patterns, trends, relationships, and significant findings.

    Data Visualization: Presenting information through charts, graphs, maps, and info graphics to make complex data easier to understand.

    Data Scraping: Automatically collecting information from websites or extracting data from documents such as PDFs.

    Statistical Analysis: Using statistical methods, such as averages, percentages, and comparisons, to interpret data accurately.

    Storytelling with Data: Using data-driven findings to develop meaningful news stories that explain important issues.

Verification and Fact-Checking: Checking the accuracy, reliability, and credibility of data before publishing a story.

Ethics and Transparency: Protecting privacy, avoiding bias, and explaining data sources and methods honestly.

Interactive Journalism: Using interactive maps, charts, and dashboards to allow readers to explore data themselves.

Accuracy and credibility are two important qualities required for data journalism. Credibility can be ensured by double checking and verifying the collected data via multiple reliable sources. Accuracy can ensured through data filtering, data cleaning, and data organization.

24. Introduction to data journalism – scope of data journalism – importance of storytelling in data journalism – types of data (primary, secondary etc.) – combination of creativity, journalistic writing and data visualization – scope of data visualization – Using multiple visualization techniques to hook the audience attention – Info graphics, bar diagram, tables, pie diagram etc.
25. Simplifying complex data is an important part of data journalism because it helps readers understand complicated information easily while preserving its accuracy and original meaning.
    The major effects of data simplification are as follows: Improves understanding – Increases accessibility – supports better storytelling – maintains accuracy – Prevents - misleading information – Saves time – builds public trust

    Simplifying complex data makes news more accessible, meaningful, and engaging. However, clarity should never come at the cost of accuracy. Journalists must simplify the presentation without distorting the original meaning or hiding important information.
26. Data journalism is the practice of collecting, analyzing, and presenting data to investigate issues and communicate news stories. It plays an important role in uncovering social problems, exposing inequalities, and holding governments and institutions accountable.
    Poverty: Data journalism helps identify poverty levels, income inequality, and differences in living standards across regions. Journalists analyze census data, household surveys, and government reports to reveal which communities lack basic necessities.
    Unemployment: It helps investigate unemployment rates, job opportunities, wage differences, and employment trends. By comparing data across age groups, genders, and regions, journalists can identify groups most affected by unemployment.

Education: Data journalism examines literacy rates, school dropout rates, student performance, teacher shortages, and access to educational facilities. It can reveal inequalities between rural and urban schools and between different social groups.
Crime: Journalists use crime statistics to identify crime trends, high-risk areas, and changes in reported offences. Comparing police records with other sources can help investigate public safety issues. However, differences in reporting practices must be considered.
Public health: Data journalism analyzes disease rates, hospital facilities, vaccination coverage, mortality rates, and access to healthcare. It helps reveal gaps in healthcare services and evaluate the effectiveness of public health programmes.

By examining public records, budgets, and official statistics, journalists can identify misuse of public funds, ineffective policies, and failures in public services. This encourages transparency and better governance.
Charts, graphs, maps, and interactive visualizations make complex social problems easier to understand. They help readers recognize the scale of an issue and make informed decisions.

[15x2=30]